

Extending works by Gunningham–Jordan–Vazirani and Kinnear, we present with J-V

The GL_N Skeins Partition Function for $M_\gamma = T^2 \times_\gamma S^1$

Definition

The *Skeins Partition Function* of a 3-manifold M is

$$\mathcal{Z}_M(t) := 1 + \sum_{N=1}^{\infty} \dim \mathrm{Sk}_{\mathrm{GL}_N}(M) \cdot t^N.$$

Theorem

For $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, the *Skeins Partition Function* of M_γ admits the following expansion:

$$\mathcal{Z}_{M_\gamma}(t) = \prod_{k=1}^{\infty} (1 - t^k)^{-c_k}, \text{ where } c_k = \frac{1}{k} \sum_{d|k} \phi\left(\frac{k}{d}\right) |\mathrm{tr}(\gamma^d) - 2| \text{ for generic } \gamma.$$

$$\text{E.g., for } \gamma = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}, \mathcal{Z}_{M_\gamma}(t) = 1 + 2t + 10t^2 + 36t^3 + \dots = (1 - t)^{-2}(1 - t^2)^{-7}(1 - t^3)^{-18} \dots$$

Want to know more? Please talk to us!

Julia Bierent

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Matthias Vancraeynest

Also interested in:
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